

Optimization

1.1 Optimization problems and modelling

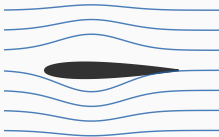
José Firmino Aguilar Madeira

Definition and general structure

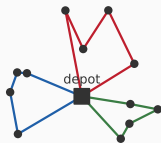
Which is the best?

To optimize is to choose the best feasible alternative.

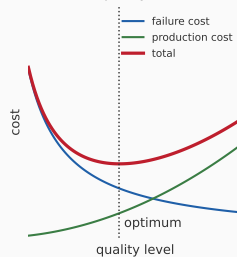
shape that minimizes drag



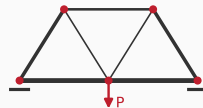
minimum-cost routes



cost / quality trade-off



minimum-weight structure

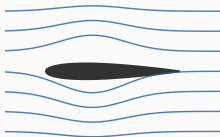


Which is the best?

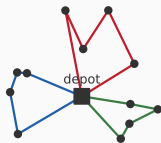
To optimize is to **choose the best feasible alternative**.

But what does **best** mean? And which alternatives are **feasible**?

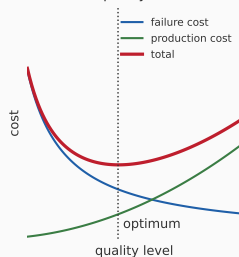
shape that minimizes drag



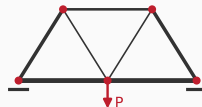
minimum-cost routes



cost / quality trade-off



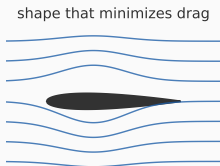
minimum-weight structure



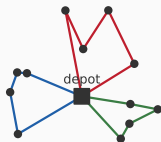
Which is the best?

To optimize is to **choose the best feasible alternative**.

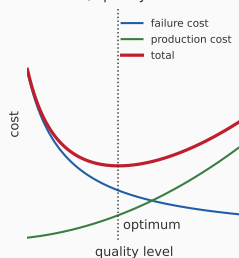
But what does **best** mean? And which alternatives are **feasible**?



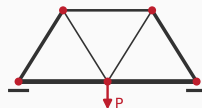
minimum-cost routes



cost / quality trade-off



minimum-weight structure



In every case there are **decisions** to make, a **criterion** to compare them and **limits** that cannot be exceeded.

In this lecture

1. What is an optimization problem?
2. How is a real problem turned into a mathematical model?
3. What types of problems are there?
4. How do we choose a method to solve it?

A concrete problem: the Gira bicycles

There are N bicycles and n stations, station i with C_i docks. How many bicycles to place at each station?

A concrete problem: the Gira bicycles

There are N bicycles and n stations, station i with C_i docks. How many bicycles to place at each station?

- **What we decide:** x_i = number of bicycles at station i , $i = 1, \dots, n$.

A concrete problem: the Gira bicycles

There are N bicycles and n stations, station i with C_i docks. How many bicycles to place at each station?

- **What we decide:** x_i = number of bicycles at station i , $i = 1, \dots, n$.
- **What “best” means:** fewer service failures — requests at an empty station, returns at a full station — estimated from the forecast demand at each station.

A concrete problem: the Gira bicycles

There are N bicycles and n stations, station i with C_i docks. How many bicycles to place at each station?

- **What we decide:** x_i = number of bicycles at station i , $i = 1, \dots, n$.
- **What “best” means:** fewer service failures — requests at an empty station, returns at a full station — estimated from the forecast demand at each station.
- **What is allowed:** $x_1 + \dots + x_n = N$ (there are no more bicycles); $0 \leq x_i \leq C_i$ (docks); x_i integer.

A concrete problem: the Gira bicycles

There are N bicycles and n stations, station i with C_i docks. How many bicycles to place at each station?

- **What we decide:** x_i = number of bicycles at station i , $i = 1, \dots, n$.
- **What “best” means:** fewer service failures — requests at an empty station, returns at a full station — estimated from the forecast demand at each station.
- **What is allowed:** $x_1 + \dots + x_n = N$ (there are no more bicycles); $0 \leq x_i \leq C_i$ (docks); x_i integer.

This is already an optimization problem.

Simplified version, for the lecture; the real work also includes redistribution throughout the day.

The same structure in other real problems



Gira shared bicycles

Câmara Municipal de Lisboa

How many bicycles to place at each station, and how to redistribute them, to minimize empty or full stations. (Project by a student of the course.)



GALP/ENACOL maritime logistics

Cabo Verde

Configuration and capacity of the ships, and logistics of fuel distribution between islands, minimizing costs and supply shortages.



Bright Optimizer

international student competition

Structural optimization: design a minimum-weight structure that withstands the imposed loads.

brightoptimizer.com

How much is optimizing worth?

Published results of operations research studies (Hillier & Lieberman), annual savings:

Continental Airlines

crew reassignment

40 M\$

Samsung Electronics

manufacturing times and inventory

200 M\$

China

large energy projects

425 M\$

Others in the same source: water management in the Netherlands (15 M\$), Citgo refineries (70 M\$), San Francisco police patrols (11 M\$).

The gain comes from **deciding better**, not from producing more:
the same machines, people and routes, organized differently.

What is an optimization problem

An **optimization problem** consists of finding the best values for a set of decision variables so as to:

- **minimize** or **maximize** an objective function;
- satisfy a set of constraints.

In the real problem	In the model
what we decide	→ decision variables $\mathbf{x} = (x_1, \dots, x_n)$
what “best” means	→ objective function $f(\mathbf{x})$: cost, profit, weight, ...
“is this allowed?”	→ constraints : conditions the solution must satisfy

For the bicycles: x_i , the service failures, and $\sum_i x_i = N$, $0 \leq x_i \leq C_i$.

Constraints

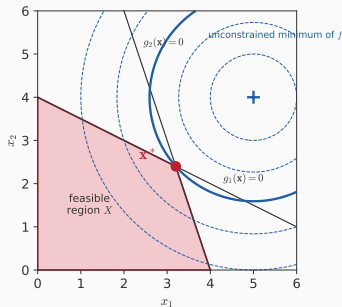
- **Inequality:** $g_j(\mathbf{x}) \geq 0$, $j = 1, \dots, J$ — limits that cannot be exceeded (capacity, maximum stress, budget). A limit “ $c(\mathbf{x}) \leq b$ ” is written $g(\mathbf{x}) = b - c(\mathbf{x}) \geq 0$.
- **Equality:** $h_\ell(\mathbf{x}) = 0$, $\ell = 1, \dots, K$ — conditions that must be satisfied exactly (mass balance, demand to be met, fixed volume).
- **Variable bounds:** $x_i^{(L)} \leq x_i \leq x_i^{(U)}$, $i = 1, \dots, n$ — the physically admissible range of each variable.

Course convention, used in all chapters: inequalities in the form $g_j(\mathbf{x}) \geq 0$ ($g \leq 0 \iff -g \geq 0$) and always *minimize* ($\max f \iff \min(-f)$).

General formulation

$$\begin{aligned} & \underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} && f(\mathbf{x}) \\ & \text{subject to} && g_j(\mathbf{x}) \geq 0, && j = 1, \dots, J \\ & && h_\ell(\mathbf{x}) = 0, && \ell = 1, \dots, K \\ & && x_i^{(L)} \leq x_i \leq x_i^{(U)}, && i = 1, \dots, n \end{aligned}$$

- **Feasible region** X : set of the $\mathbf{x}$ that satisfy all the constraints.
- **Optimal solution** $\mathbf{x}^* \in X$: $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in X$.
- A constraint is called **active** at $\mathbf{x}^*$ if it holds there with equality.

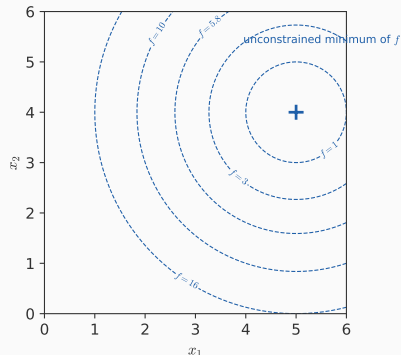


In the figure, g_1 and g_2 are both active at $\mathbf{x}^*$.

The geometry of the problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) = (x_1 - 5)^2 + (x_2 - 4)^2 \\ \text{s.t.} \quad & g_1 = 8 - x_1 - 2x_2 \geq 0, \quad g_2 = 12 - 3x_1 - x_2 \geq 0, \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

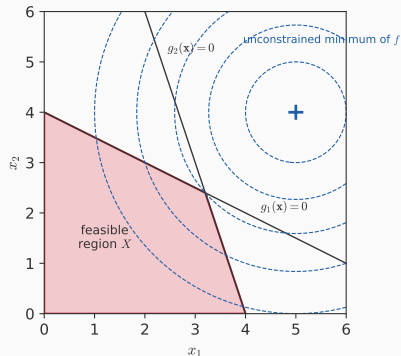
1. Level curves of f : without constraints, the minimum would be $(5, 4)$.



The geometry of the problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) = (x_1 - 5)^2 + (x_2 - 4)^2 \\ \text{s.t.} \quad & g_1 = 8 - x_1 - 2x_2 \geq 0, \quad g_2 = 12 - 3x_1 - x_2 \geq 0, \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

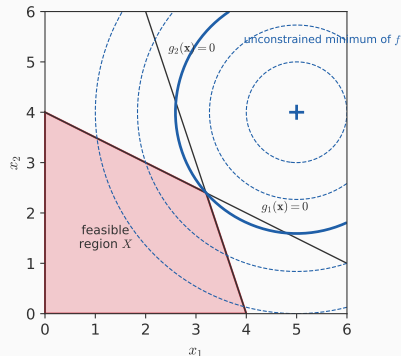
1. Level curves of f : without constraints, the minimum would be $(5, 4)$.
2. Feasible region X : **we can only choose here** — and $(5, 4) \notin X$.



The geometry of the problem

$$\begin{aligned} \min \quad & f(\mathbf{x}) = (x_1 - 5)^2 + (x_2 - 4)^2 \\ \text{s.t.} \quad & g_1 = 8 - x_1 - 2x_2 \geq 0, \quad g_2 = 12 - 3x_1 - x_2 \geq 0, \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

1. Level curves of f : without constraints, the minimum would be $(5, 4)$.
2. Feasible region X : **we can only choose here** — and $(5, 4) \notin X$.
3. Go down the level curves to the lowest one that still **touches** X .

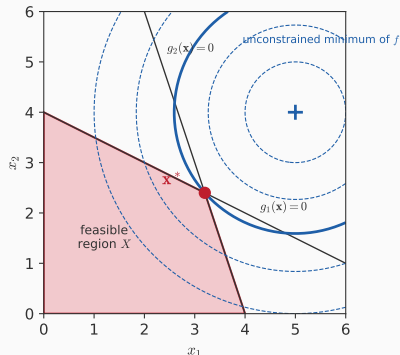


The geometry of the problem

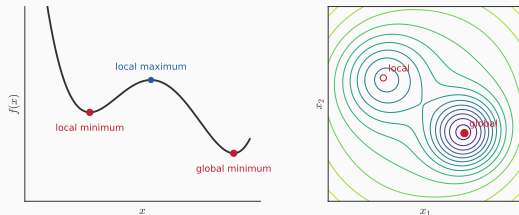
$$\begin{aligned} \min \quad & f(\mathbf{x}) = (x_1 - 5)^2 + (x_2 - 4)^2 \\ \text{s.t.} \quad & g_1 = 8 - x_1 - 2x_2 \geq 0, \quad g_2 = 12 - 3x_1 - x_2 \geq 0, \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

1. Level curves of f : without constraints, the minimum would be $(5, 4)$.
2. Feasible region X : **we can only choose here** — and $(5, 4) \notin X$.
3. Go down the level curves to the lowest one that still **touches** X .
4. The contact: $\mathbf{x}^* = (3.2, 2.4)$, $f^* = 5.8$, with g_1 , g_2 active.

The optimum is the minimum of f among the feasible solutions — not necessarily the unconstrained minimum. This idea returns in 4.1–4.2 (Lagrange, KKT) and in 6.2 (Pareto).



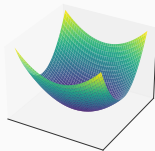
Local minimum and global minimum



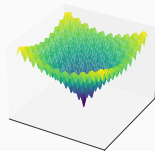
- $\mathbf{x}^* \in X$ is a **global minimum** if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in X$.
- $\mathbf{x}^* \in X$ is a **local minimum** if there exists $\delta > 0$ with $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in X$ with $\|\mathbf{x} - \mathbf{x}^*\| < \delta$ (**strict**: $<$ for $\mathbf{x} \neq \mathbf{x}^*$). Every global minimum is local.
- Most methods (Chs. 2–4) look for local minima (strictly speaking, in the best case and under assumptions, stationary/KKT points); global optimization methods (Ch. 5) try to escape them. If the problem is *convex*, every local minimum is global.

Unimodal and multimodal functions

unimodal: $f = 0.5x_1^2 + x_2^2$



multimodal: Ackley function



- **Unimodal** (rigorous definition in 2.1): a single local minimum, which is global; the descent methods of Chs. 2–3 find it, under the assumptions of each.
- **Multimodal**: several local minima; the result depends on the initial point. Motivates global optimization methods (Ch. 5).

Weierstrass theorem

A continuous function on a closed and bounded (compact) set has a maximum and a minimum on that set — it guarantees that a solution exists, it does not say how to find it.

Classification of problems

Constraints	unconstrained	constrained	
Functions	linear (LP)	nonlinear	
Variables	continuous	discrete / integer	mixed
Objectives	single objective	multiobjective (Ch. 6)	
Information	with derivatives	derivative-free	

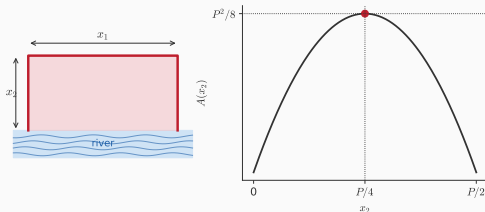
The theoretical core of the course is **continuous** nonlinear optimization, constrained and unconstrained, with one or several objectives; Ch. 5 also shows how some metaheuristics extend naturally to discrete and combinatorial variables.

Examples of optimization problems

Example 1 — fence by the river

Maximize the area of a rectangular enclosure with one side against a river (no fence on that side), with P metres of fence available.

- **Variables:** x_1 (length), x_2 (width)
- **Objective:** maximize $A(x_1, x_2) = x_1 x_2$
- **Constraints:** $x_1 + 2x_2 \leq P$; $x_1, x_2 \geq 0$



Solution

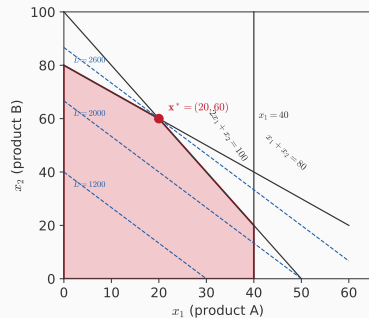
All the fence will be used, so $x_1 = P - 2x_2$ and $A(x_2) = Px_2 - 2x_2^2$. From $A'(x_2) = P - 4x_2 = 0$ we get $x_2^* = P/4$, $x_1^* = P/2$ and $A_{\max} = P^2/8$ ($A'' = -4 < 0$: it is a maximum).

Example 2 — production plan

A company makes two products with unit profit 40 € (A) and 30 € (B). Per unit, A uses 2 h of machine time and 1 h of labour; B uses 1 h of each. There are 100 h of machine time, 80 h of labour, and the demand for A does not exceed 40 units.

$$\begin{aligned} \max \quad & L = 40x_1 + 30x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 100 \quad (\text{machine}) \\ & x_1 + x_2 \leq 80 \quad (\text{labour}) \\ & x_1 \leq 40 \quad (\text{demand}), \quad x_1, x_2 \geq 0 \end{aligned}$$

Linear problem: the feasible region is a polygon and the optimum is at a vertex. The level lines $L = \text{const}$ slide up to the last feasible point: $\mathbf{x}^* = (20, 60)$, $L^* = 2600$ €. Active constraints: machine and labour.



Example 3 — can of given volume: modelling

Make a cylindrical can with $V = 330 \text{ cm}^3$ using the least material.

Before seeing the answer: what are (1) the variables, (2) the objective, (3) the constraints?

Example 3 — can of given volume: modelling

Make a cylindrical can with $V = 330 \text{ cm}^3$ using the least material.

Before seeing the answer: what are (1) the variables, (2) the objective, (3) the constraints?

- **Variables:** the radius r and the height h .

Example 3 — can of given volume: modelling

Make a cylindrical can with $V = 330 \text{ cm}^3$ using the least material.

Before seeing the answer: what are (1) the variables, (2) the objective, (3) the constraints?

- **Variables:** the radius r and the height h .
- **Objective:** minimize the sheet area, $A(r, h) = 2\pi r^2 + 2\pi rh$ (bottom, lid and side).

Example 3 — can of given volume: modelling

Make a cylindrical can with $V = 330 \text{ cm}^3$ using the least material.

Before seeing the answer: what are (1) the variables, (2) the objective, (3) the constraints?

- **Variables:** the radius r and the height h .
- **Objective:** minimize the sheet area, $A(r, h) = 2\pi r^2 + 2\pi rh$ (bottom, lid and side).
- **Constraints:** $\pi r^2 h = V$ (the volume is given); $r, h > 0$.

Example 3 — can of given volume: modelling

Make a cylindrical can with $V = 330 \text{ cm}^3$ using the least material.

Before seeing the answer: what are (1) the variables, (2) the objective, (3) the constraints?

- **Variables:** the radius r and the height h .
- **Objective:** minimize the sheet area, $A(r, h) = 2\pi r^2 + 2\pi rh$ (bottom, lid and side).
- **Constraints:** $\pi r^2 h = V$ (the volume is given); $r, h > 0$.

$$\min_{r, h} 2\pi r^2 + 2\pi rh \quad \text{s.t.} \quad \pi r^2 h = V, \quad r, h > 0$$

Example 3 — can of given volume: solving

Nonlinear problem with one equality constraint.

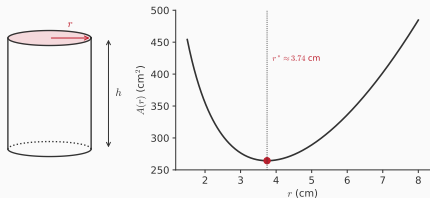
Eliminating $h = V/(\pi r^2)$:

$$A(r) = 2\pi r^2 + \frac{2V}{r}, \quad A'(r) = 4\pi r - \frac{2V}{r^2} = 0$$

$$\Rightarrow r^* = \sqrt[3]{V/(2\pi)} \approx 3.74 \text{ cm.}$$

$$A''(r) = 4\pi + 4V/r^3 > 0 \Rightarrow \text{minimum;}$$

$A^* \approx 264.4 \text{ cm}^2$. Hence $h^* = 2r^* \approx 7.5 \text{ cm}$: the optimal can is as tall as it is wide (real ones are not — there are other costs and constraints outside this model).



What the examples have in common

	Fence	Production	Can
Variables	x_1, x_2	x_1, x_2	r, h
Objective	$\max x_1 x_2$	$\max 40x_1 + 30x_2$	$\min 2\pi r^2 + 2\pi rh$
Constraints	$x_1 + 2x_2 \leq P$	3 inequalities	$\pi r^2 h = V$
Type	nonlinear	linear	nonlinear
Solution	eliminate + differentiate	graphical (vertex)	eliminate + differentiate

With more variables, or functions that cannot be differentiated by hand, these tricks no longer work: that is what the **numerical methods** of the following chapters are for.

Example 2 in a solver: what the computer returns

```
% max 40x1+30x2 -> min -(40x1+30x2); constraints A*x <= b
c = [-40; -30];
A = [2 1; 1 1; 1 0]; b = [100; 80; 40];
lb = [0; 0];
[x, L, ~, ~, lambda] = linprog(c, A, b, [], [], lb);
x, -L % x = (20, 60), -L = 2600
lambda.ineqlin % shadow prices: 10, 20, 0
```

- The **shadow prices**: an extra hour of machine time is worth 10 €, one of labour 20 €; the demand constraint, inactive, is worth 0. This is the information the solver gives *besides* the solution — 4.2.
- Check by hand: in the two active constraints,
 $2u_1 + u_2 = 40$ and $u_1 + u_2 = 30$ give $u_1 = 10$, $u_2 = 20$.

THE SAME IN PYTHON

```
from scipy.optimize import linprog
res = linprog(c=[-40,-30],
              A_ub=[[2,1],[1,1],[1,0]],
              b_ub=[100,80,40], bounds=[(0,None)]*2)
res.x, -res.fun # (20,60), 2600
res.ineqlin.marginals # -10, -20, 0
```

The same mathematical formulation in two languages; the internal solver may differ. SciPy's marginals come with a negative sign by convention (minimization of $-L$): -10 and -20 are MATLAB's 10 and 20.

Mathematical modelling of problems

Two different questions

1. What is the problem? —
modelling

reality $\longrightarrow \mathbf{x}, f, g_j, h_\ell$

2. How do we solve it? —
optimization

$(\mathbf{x}, f, g_j, h_\ell) \longrightarrow \text{algorithm} \longrightarrow \mathbf{x}^*$

An excellent algorithm applied to the wrong model
solves **the wrong problem** very well.

This course is mostly about question 2; but each chapter starts with question 1.

From the real situation to the model

1. **Identify the decision variables.** What can we choose? Quantities, dimensions, positions, yes/no. Fix the units.
2. **Formulate the objective function.** A single metric, in consistent units, written as a function of the variables. If there are several criteria: aggregate them or treat as multiobjective (Ch. 6).
3. **Write the constraints.** Limited resources, physical laws, demand, safety, signs of the variables. Check that each one is expressed only in terms of the variables and known data.

Two common mistakes

Treating a *datum* as a variable (or vice versa); adding in the objective function quantities with different units (euros with hours).

Complete example — transportation network

A company distributes goods from warehouses $i \in \{1, \dots, I\}$ to customers $c \in \{1, \dots, C\}$. Each warehouse has supply s_i , each customer demands d_c , and shipping one unit from i to c costs p_{ic} (*data*).

Step 1 — variables

$x_{ic} \geq 0$: quantity shipped from i to c .

Step 2 — objective

$\min \sum_{i=1}^I \sum_{c=1}^C p_{ic} x_{ic}$ (total cost, in €)

Step 3 — constraints

$\sum_c x_{ic} \leq s_i$ (do not exceed supply), $\sum_i x_{ic} = d_c$ (meet demand), $x_{ic} \leq q_{ic}$ (route capacity).

Complete example — and after the model?

- **Check** whether the model is feasible: $\sum_i s_i \geq \sum_c d_c$ is a necessary condition (sufficient only without route capacities).
- **Classify**: linear, continuous, constrained — solved with the simplex method or interior points, even with thousands of variables.
- **Solve** numerically and **interpret**: which constraints are active? How much is relaxing each one worth (shadow prices)?
- **Validate** with real data and refine the model (delivery times, integer vehicles, several commodities, ...).

A model is always a simplification; the quality of the decision depends more on the model than on the algorithm that solves it.

OPTIMIZATION IN PRACTICE — maritime fuel logistics between islands (GALP/ENACOL, Cabo Verde)

Decision which ships to use (and with what capacity), which routes and schedule between the islands, and how much of each fuel to unload at each port.

Variables quantities per trip, port and product (continuous) *and* choices of ship, route and sequence of ports (discrete).

Objective minimize the distribution cost and supply shortages.

Constraints demand of each island, capacity of the onshore tanks and of the ships, port conditions, deadlines.

Why it is hard it is the transportation network of the previous pages with integer decisions, several products and the time dimension: the result is not a solution, it is a *decision support system* that is used again whenever the data change.

Methods mixed(-integer) linear programming models, integrated into a decision support system. The formulation — what is a variable, a datum or a constraint — is the decisive part.

Textbook problem vs. real problem

	fence, production, can	real problems (e.g. wind farm, structure)
is f known?	explicit formula	comes from a simulation, a finite element model or data
cost of evaluating f	negligible	seconds to hours per evaluation
derivatives	by hand, exact	often unavailable or unreliable
variables	2, continuous	tens to thousands; continuous, integer, mixed
constraints	1–3, simple	many, some only checkable a posteriori
objectives	one	often several, conflicting
noise	none	measurements, stochastic simulations

Textbook examples help us understand the methods. Real problems explain **why there is a scientific field called optimization**: almost every row of the right-hand column is a reason for a chapter of this course; integer variables and noise lie outside the core.

Course overview

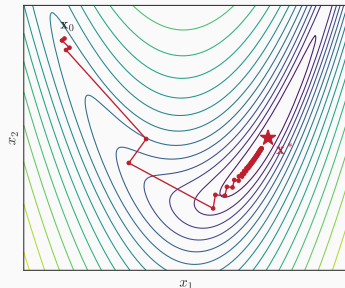
How it is solved numerically

Almost all methods are **iterative**:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{S}_k, \quad k = 0, 1, 2, \dots$$

- $\mathbf{x}_0$: initial point; $\mathbf{x}_k$: the point after k iterations
- $\mathbf{S}_k$: search direction (Ch. 3)
- $\lambda_k > 0$: step length along the direction (Ch. 2)
- stop when $\|\nabla f(\mathbf{x}_k)\|$, the step or the change in f are small

The following chapters differ in *how* they choose $\mathbf{S}_k$ and λ_k , in what information about f they use, and in how they handle the constraints.



Iterations of a steepest descent method on the level curves of f .

The question that runs through the course

What do we know about f ? What does it cost to obtain? What can we guarantee with it?

	information on f	typical cost	what is guaranteed
values of f only (order 0)	comparisons $f(\mathbf{x}) < f(\mathbf{x}')$	many evaluations (n_f)	NM: no general guarantee; formal direct search (e.g. GLODS, 5.1): stationarity under suitable assumptions
gradient (order 1)	where f decreases	$n_g = 1$ (analytical) or $2n$ evaluations of f (central)	stationarity in the limit, from any $\mathbf{x}_0$ (under regularity + suitable line search; 3.3.1)
Hessian (order 2)	curvature	$n_H = 1$ (analytical) or $O(n^2)$ evaluations of f	fast convergence, near the solution
structure (linear, convex)	global shape of f and X	—	global optimum

We count separately n_f (evaluations of f), n_g (of f' ; in $\mathbb{R}^n$, gradients) and n_H (of f'' ; Hessians). Each chapter gives a different trade-off between information, cost and guarantee. Before solving we must decide *what the variables are, what “best” means, what is allowed and what it costs to evaluate a decision*.

How to choose a method: questions before the algorithm

Question	Ch.	Methods
One variable?	2	golden section, successive quadratic interpolation (Powell's method), bisection, Newton–Raphson
Several variables, unconstrained?	3	derivative-free: random and grid search, Nelder–Mead, Box, Hooke–Jeeves; with derivatives: steepest descent, Newton, conjugate gradients
Any constraints?	4	Lagrange, KKT, exterior and interior penalty
Several local minima?	5	GLODS, simulated annealing, genetic algorithms
Several objectives?	6	Pareto, aggregation, ε -constraint, DMS, NSGA-II
Do we have derivatives?	2–3	with or without derivatives (order 0, 1, 2)
Is f expensive or a black box?	all	the key resource is now the number of evaluations of f

There is no “best optimization algorithm”.

Only the method suited to the available information and to the structure of the problem.

Next (1.2): where optimization is used.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 2024 Release, McGraw-Hill, 2024 — case studies cited in this deck.
- Arora, J. S., *Introduction to Optimum Design*, 3rd ed., Academic Press, 2011 — Ch. 2, formulation.

Resources

- MATLAB: `linprog` (Optimization Toolbox), shadow prices in `lambda.ineqlin`.
- Python: `scipy.optimize.linprog`, shadow prices in `res.ineqlin.marginals`. Link: brihtoptimizer.com.